



MODIFIED PREDATOR-PREY MODEL

DEPARTMENT OF MATHEMATICS, BITS PILANI GOA CAMPUS

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ABSTRACT

This paper explores the mathematical aspects of ecological systems, focusing on how populations interact and influence each other's growth. It modifies the predator-prey model, examining a case with two types of preys and two types of predators. Our model is inspired from a real-life food cycle chain of Lion-deer-grass. Lion is the ultimate-predator, Deer is a predator for grass but prey for lion. Grass is prey for deer. The three-dimensional Lotka-Volterra model is analyzed with an additional assumption, categorizing its solutions into three stages to illustrate the dynamics between prey and their predators. Notably, when middle level predators are absent, primary prey population grows exponentially and top-level predator decays exponentially, as expected biologically. However, absence of primary prey leads to the extinction of all populations and absence of top-level predator will result in classic Lotka Volterra model. Stability analysis is conducted by calculating eigenvalues of the Jacobian matrix, and the relationships between species are visually represented using MATLAB Software.

INTRODUCTION

The Lotka model, born in 1921, marked the dawn of species interaction study, using differential equations to depict parasite-prey dynamics. Volterra expanded on this, detailing the relationship further. This predator-prey model illustrates how species coexist, their populations shaped by births, deaths, and interactions. Lotka-Volterra was groundbreaking, introducing multi-species dynamics in a shared ecosystem. This model, adaptable to various dimensions, embodies the intricate dance of predator and prey. Equations describing the species' relationships and equilibrium points are derived, revealing insights into their interactions.

PREDATOR PREY MODEL

This model illustrates a linear three-species food chain, where the prey 'x' is consumed by a middle-level species 'y'. Subsequently, species 'y' becomes the prey of a top-level predator 'z'. Examples of such three-species ecosystems include scenarios like mouse-snake-owl interactions.

ASSUMPTIONS

- 1) The prey population is assumed to have an unlimited food supply.
- 2) Interactions between predators and prey are proportional to the product of their populations.
- 3) Predators rely on prey as their primary source of sustenance.

Equations of 2 – D Lotka Volterra

$$\frac{dx}{dt} = ax - bxy$$

$$\frac{dy}{dt} = -dy + cxy$$

Equate the equations to 0 to get stationary points

The steady states are $(0,0)$; $\left(\frac{d}{c}, \frac{a}{b}\right)$

$$J = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix}$$

$$J = \begin{bmatrix} a - by & -bx \\ cy & cx - d \end{bmatrix}$$

We will check the stability of (0,0)

$$J_{(0,0)} = \begin{bmatrix} \mathbf{a} & \mathbf{0} \\ \mathbf{0} & -\mathbf{d} \end{bmatrix}$$

The eigen values of resultant matrix are $(\lambda = a)$ or $(\lambda = -d)$

(0,0) is unstable because both the eigen values have different signs.

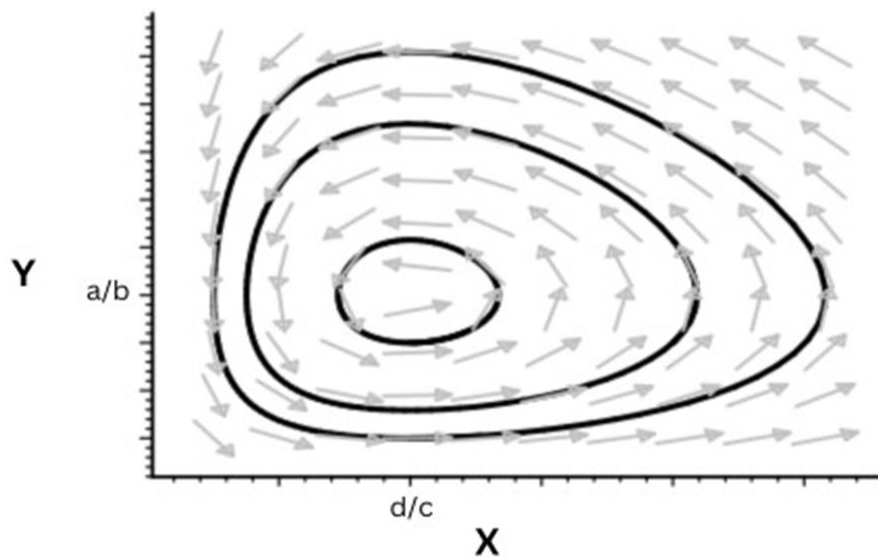
Now we will check the stability around $\left(\frac{d}{c}, \frac{a}{b}\right)$

$$J_{\left(\frac{d}{c}, \frac{a}{b}\right)} = \begin{bmatrix} \mathbf{0} & -\frac{\mathbf{bd}}{\mathbf{c}} \\ \frac{\mathbf{ca}}{\mathbf{b}} & \mathbf{0} \end{bmatrix}$$

The eigen values of resultant matrix are $\lambda = \pm i\sqrt{ad}$

X-Null Cline: $y = \frac{a}{b}$

Y-Null Cline: $x = \frac{d}{c}$



PREDATOR PREY MODEL IN 3-DIMENSIONS

In the 3D Lotka-Volterra system, the addition of the new species 'z' represents predators that rely solely on the middle-level population 'y' for sustenance.

The modified Lotka-Volterra model can be written as

$$\frac{dx}{dt} = ax - bxy$$

$$\frac{dy}{dt} = -cy + dxy - eyz$$

$$\frac{dz}{dt} = -fz + gyz$$

a is the growth rate of the prey in the absence of predators

b represent the effects of the predation on the preys

c referred to the natural death rate of the predator in the absence of prey

e denotes the effect of predation on species y by species z

f denotes the natural death rate of the predator z in the absence of prey,

g is the efficiency and propagation rate of the predator z in the presence of prey

Equate the equations to 0 to find stationary points:

$$(x = 0) \text{ or } (a - by = 0)$$

$$(y = 0) \text{ or } (-c + dx - ez = 0)$$

$$(z = 0) \text{ or } (-f + gy = 0)$$

If $x \neq 0$ and (if $ga \neq bf$) Steady state is $\left(\frac{c}{d}, \frac{a}{b}, 0\right)$

If $z \neq 0$ and (if $ga \neq bf$) Steady state is $\left(0, \frac{f}{g}, \frac{-c}{e}\right)$

Thus $\left(0, \frac{f}{g}, \frac{-c}{e}\right)$ will be rejected because of our given condition i.e.
 $x, y, z \geq 0$

Final Steady states: $\left(\frac{c}{d}, \frac{a}{b}, 0\right); (0,0,0)$

STAGES:

CASE-1($x=0$):

In the absence of first prey, the model will be as follows:

$$\frac{dx}{dt} = 0$$

$$\frac{dy}{dt} = -cy - eyz$$

$$\frac{dz}{dt} = -fz + gyz$$

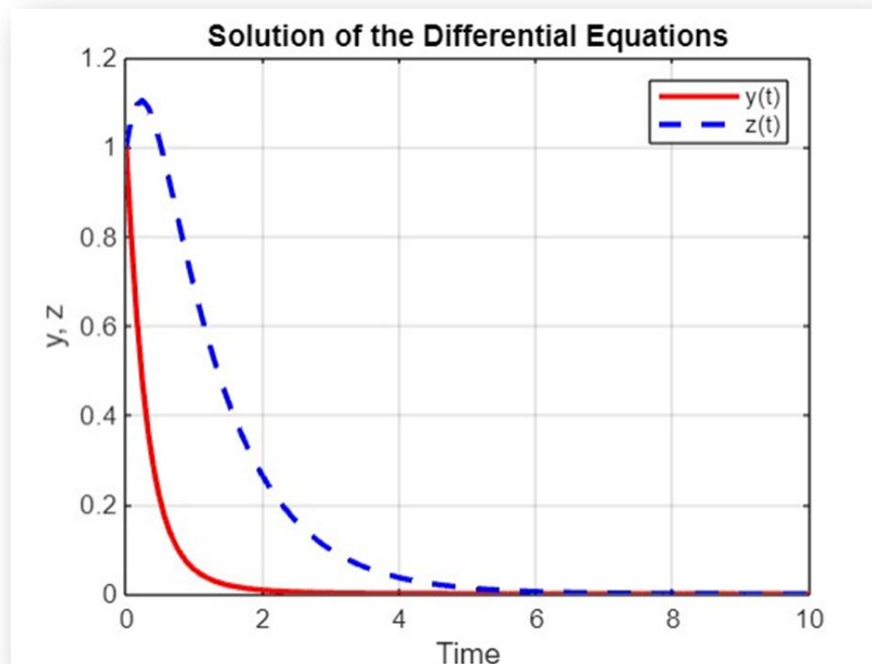
$$\frac{dy}{dt} = -cy - eyz \quad \Rightarrow \quad y = y_0 e^{-ct} e^{-ezt}$$

$$0 \leq z < \infty \Rightarrow 0 \leq e^{-ezt} \leq 1 \quad \text{Putting } t \rightarrow \infty$$

$$\Rightarrow y \rightarrow 0$$

$$\frac{dz}{dt} = -fz + gyz \quad \Rightarrow \quad z = z_0 e^{-ft} e^{gyt}$$

$$t \rightarrow \infty \Rightarrow z \rightarrow 0$$



Analysis of Case 1

When the population of prey is zero on the $x=0$ plane, the system simplifies to the classical Lotka-Volterra model. As confirmed by the integrals above, if $x=0$, the population of $y(t)$ dwindles due to the absence of its prey. Consequently, without its food source, the population of $z(t)$ also faces extinction. Therefore, without $x(t)$, both predator populations will eventually perish.

CODE FOR THE ABOVE GRAPH- mentioned in the end

CASE-2($y=0$):

In the absence of second prey, the model will be as follows:

$$\frac{dx}{dt} = ax$$

$$\frac{dy}{dt} = 0$$

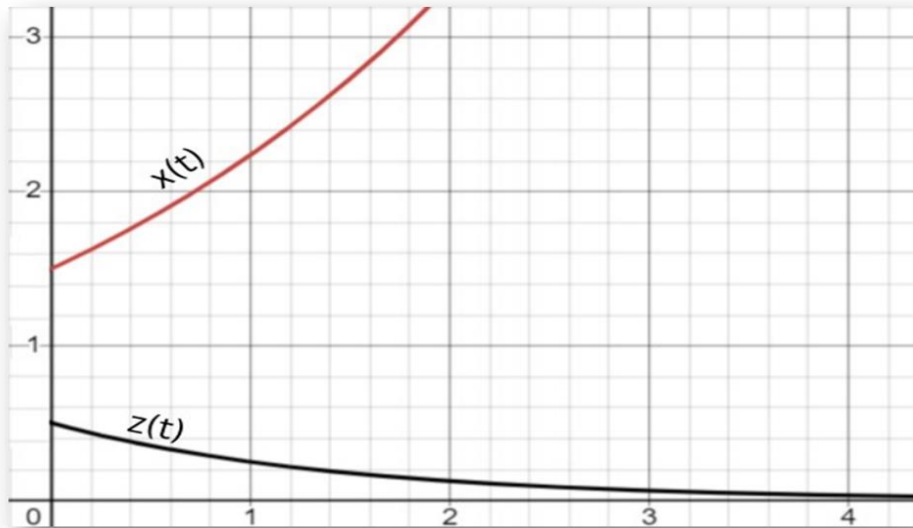
$$\frac{dz}{dt} = -fz$$

$$\frac{dx}{dt} = ax \quad \Rightarrow \quad x = x_0 \cdot e^{at}$$

$$t \rightarrow \infty \Rightarrow x \rightarrow \infty$$

$$\frac{dz}{dt} = -fz \quad \Rightarrow \quad z = z_0 \cdot e^{-ft}$$

$$t \rightarrow \infty \Rightarrow z \rightarrow 0$$



Analysis of Case-2

Yet, on the $y=0$ plane, the prey population experiences exponential growth in the absence of predation. Consequently, the population of $z(t)$ lacks a food source and thus faces extinction. Conversely, the prey population $x(t)$ undergoes a contrasting phenomenon because there is no interaction between $x(t)$ and $z(t)$.

CASE 3

In the absence of last predator, the model will be as follows:

$$\frac{dx}{dt} = ax - bxy$$

$$\frac{dy}{dt} = -cy + dxy$$

$$\frac{dz}{dt} = 0$$

Thus the model turned into classic LotkaVolterra predator prey model and The solving is just like we did in 2-D.

Analysis of Case 3

However, on the $z=0$ plane, the system simplifies entirely to the classical Lotka-Volterra model. Here, the predator population increases with prey and decreases in its absence.

EQUILIBRIUM STATES:

Equilibrium points in a model refer to stable states where the system remains unchanged over time. These points represent a balance between various factors or variables within the model.

$$J = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} & \frac{\partial g}{\partial z} \\ \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} & \frac{\partial h}{\partial z} \end{bmatrix}$$

Steady states: $\left(\frac{c}{d}, \frac{a}{b}, 0\right), (0, 0, 0)$

$$J = \begin{bmatrix} a - by & -bx & 0 \\ dy & -c + dx - ez & -ey \\ 0 & gz & -f + gy \end{bmatrix}$$

We will check stability at $(0,0,0)$

$$J_{(0,0,0)} = \begin{bmatrix} a & 0 & 0 \\ 0 & -c & 0 \\ 0 & 0 & -f \end{bmatrix}$$

The Eigen values of the resulting matrix are

$$\lambda = a, \lambda = -c, \lambda = -f$$

Only one of the eigen values is positive and other two eigen values are negative .

So we can infer that (0,0,0) is an unstable steady state.

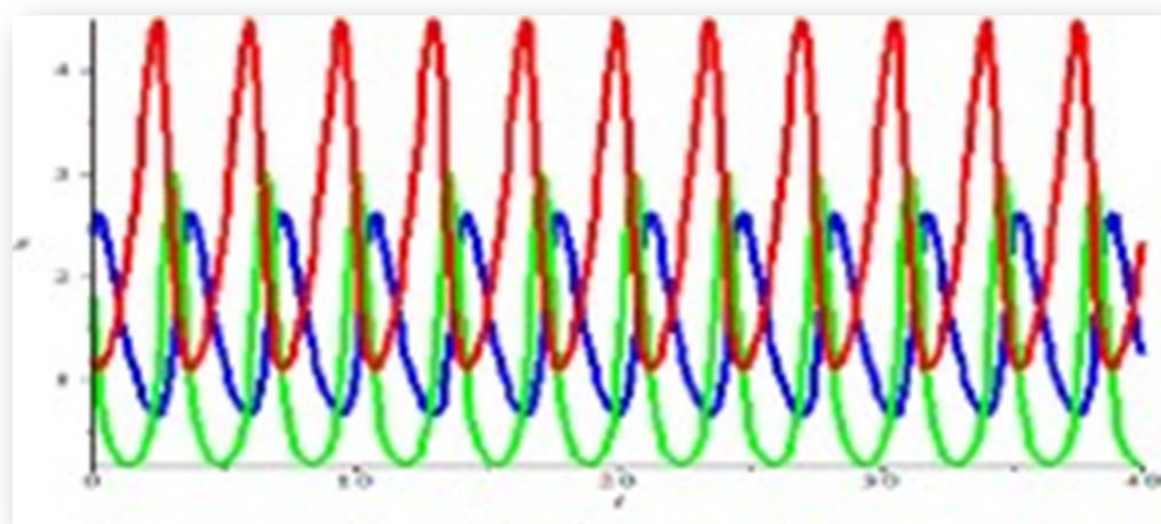
Now, we will check the stability at $\left(\frac{c}{d}, \frac{a}{b}, 0\right)$

$$J_{\left(\frac{c}{d}, \frac{a}{b}, 0\right)} = \begin{bmatrix} 0 & \frac{-bc}{d} & 0 \\ \frac{ad}{b} & -c & \frac{-ea}{b} \\ 0 & 0 & -f + \frac{ga}{b} \end{bmatrix}$$

The Eigen values of the resultant matrix are -

$$\lambda = \frac{ga}{b} - f, \lambda = \pm i\sqrt{ac}$$

If $\frac{ga}{b} - f < 0$, then the point will be stable otherwise if $\frac{ga}{b} - f \geq 0$, then the steady state will be unstable



Code for Case-1

```
% Define the parameters

c = 1;

e = 2;

f = 1;

g = 2;

% Initial conditions

y0 = 1;

z0 = 1;

z_initial = [y0; z0];

% Time span

tspan = [0 10];

% Solve the differential equations

[t, z] = ode45(@(t, z) [-c*z(1) - e*z(1)*z(2); -f*z(2) + g*z(1)*z(2)], tspan, z_initial);

% Extract y and z from the solution

y = z(:, 1);

z = z(:, 2);

% Plot the results

plot(t, y, 'r-', 'LineWidth', 2);

hold on;

plot(t, z, 'b--', 'LineWidth', 2);

xlabel('Time');

ylabel('y, z');

title('Solution of the Differential Equations');

legend('y(t)', 'z(t)');

grid on;
```

REFERENCES

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